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**Seyr-i Öneri: An Ethics-First Hybrid Retail
Recommendation System with Cultural, Cognitive,
Temporal, Causal, and Explainable Signals**

*Design and Implementation of a Culturally Grounded, Cognitively Aware,
and Causally Honest Hybrid Retail Recommendation System: Seyr-i Öneri*

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CONGRESS ACCEPTANCE, ACADEMIC DECLARATION, AND ETHICAL STATEMENT

This report is the English full-text academic and technical version of the study entitled **Seyr-i Öneri**, prepared for VI. International British Congress on Interdisciplinary Scientific Research & Practices. The English title of the accepted abstract is “Design and Implementation of a Culturally Grounded, Cognitively Aware, and Causally Honest Hybrid Retail Recommendation System: Seyr-i Öneri”. The paper was accepted under reference number **AC-06060** on **28.05.2026** for oral presentation and inclusion in the congress proceedings.

This text is not structured as a course submission document. It is prepared as an extended English full paper for an accepted international congress contribution. Accordingly, the title page, abstract, method, findings, discussion, conclusion, references, and appendices are organized to satisfy both conference full-paper expectations and the traceability requirements of a technical research report.

No real personal data were used in the study. Training, evaluation, and demonstration data were generated synthetically; the data-generation chain, experimental artefacts, and test outputs were preserved in a reproducible form. Religious and political orientation fields, which may fall under special-category personal data in GDPR Article 9 and KVKK Article 6 contexts, can exist as optional database/interface fields, but they are not used in model training or runtime inference. This decision is not left as a textual ethical claim only; it is technically enforced through `MODEL_FEATURE_COLUMNS`, automated tests, and a CI-level `ethics-guard` check.

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I also thank all faculty members who broadened my professional perspective during my software engineering education and taught me not only how to write code, but also how to define a problem, defend a system, produce evidence, and remain responsible toward users. I am grateful to my family for their encouragement, patience, and unconditional support during the long implementation and writing process. I would also like to express my sincere gratitude to my highly esteemed teacher, Cemil Baki KIYAK, for everything he has taught me.

Seyr-i Öneri is not merely a recommendation engine; it is an attempt to bring culture, attention, causality, explainability, and ethics onto the same engineering surface. For this reason, the study has also been a personal learning journey showing that software engineering must be considered together with social responsibility.

ABSTRACT

This study designs, implements, and experimentally evaluates **Seyr-i Öneri**, an ethics-first hybrid recommendation system that addresses four structural limitations frequently observed in modern retail recommendation systems within a single architecture. These limitations are: representing country only as a flat categorical label; presenting recommendation lists of fixed length to every user; confusing correlational scores with causal effect; and failing to move explainable artificial intelligence outputs to the user’s moment of decision.

The proposed system combines five scientific contributions in one open and reproducible architecture. First, *Cultural Context Embedding* (CCE) maps countries into a continuous vector space derived from Hofstede’s six cultural dimensions. Second, the *Cognitive Load Aware Recommender* (CLAR) estimates cognitive load from in-session behavioural signals and dynamically adjusts recommendation-list length, diversity coefficient, and confidence threshold. Third, *Time-Decay Behavioural Memory* (TDM) uses the Ebbinghaus exponential forgetting curve to reduce the weight of historical interactions over time. Fourth, the *Causal Uplift Re-ranker* (CUR) estimates the heterogeneous treatment effect of recommendation exposure with a T-learner approach. Fifth, *Just-in-Time Explainability* (JIT-XAI) presents SHAP-based feature contributions next to each recommendation card in a form that can be understood by the user.

The system was developed with FastAPI, Jinja2, SQLite, scikit-learn, LightGBM, SHAP, and vanilla JavaScript. On the interface side, it includes a three-dimensional Hofstede cultural globe, a live ethics and fairness dashboard, a Server-Sent Events based recommendation stream, and a Gemini-based assistant that can read page context. The data layer is based on a documented synthetic data-generation chain: a 100,000-row legacy interaction corpus, a 60,000-row natural conditional dataset, and a live demonstration database containing 119 products, 56 users, and 1002 interaction events.

Experiments show that the LightGBM model reaches 0.4510 NDCG@10 on the natural dataset in the classical category/ranking prediction setting, while the reframed binary engagement/satisfaction task achieves 0.9004 accuracy and 0.957 AUC. The ablation experiments for the five innovation layers show that CCE remains neutral in terms of engagement accuracy, CLAR reduces average top-N from 8.00 to 5.47 and thereby lowers cognitive burden, TDM measurably reduces recency strength to 0.958, CUR captures heterogeneous treatment effects with an uplift standard deviation of 0.047, and JIT-XAI generates explanations with only 4.56 ms additional p99 latency. External validation on MovieLens-100k indicates that TDM, CUR, and JIT-XAI are transferable across datasets, whereas CCE and CLAR require country and behavioural signals and therefore cannot be applied directly to every public dataset.

As a result, Seyr-i Öneri offers a defensible reference implementation beyond a single accuracy number. It contributes to cultural transfer, cognitive-load adaptation, temporal memory,

causal-effect separation, and real-time user-facing explanation. With 80 automated tests, approximately 75 percent line coverage, an ethical feature contract, reproducible experimental artefacts, and an MIT-licensed open-source structure, the study provides an internationally presentable academic and engineering foundation.

Keywords: recommender systems, hybrid recommendation, Hofstede cultural dimensions, cognitive load, causal uplift, explainable artificial intelligence, SHAP, ethical AI, GDPR, KVKK, FastAPI, LightGBM.

SHORT ABSTRACT

This study presents **Seyr-i Öneri**, an ethics-first hybrid retail recommendation system that addresses four persistent limitations of contemporary recommenders within a single reproducible software architecture: country treated as a flat categorical identifier, fixed-length recommendation lists that ignore the user’s cognitive bandwidth, correlational scoring that conflates exposure with causal impact, and explanations confined to engineering dashboards rather than the user’s point of decision.

The system integrates five interlocking contributions. Cultural Context Embedding (CCE) projects countries into Hofstede’s six-dimensional cultural space. Cognitive Load Aware Recommender (CLAR) estimates working-memory pressure from dwell variance, click velocity, hover, scroll thrash and short-bounce signals, adapting top-N length, diversity and confidence thresholds accordingly. Time-Decay Behavioural Memory (TDM) applies Ebbinghaus exponential forgetting to historical interactions. Causal Uplift Re-ranker (CUR) estimates heterogeneous treatment effects through a T-learner. Just-in-Time Explainability (JIT-XAI) renders SHAP-based feature attributions next to every recommendation card with a plain-language counterfactual explanation.

The implementation is built with FastAPI, Jinja2, SQLite, scikit-learn, LightGBM, SHAP and vanilla JavaScript. It ships with an interactive 3D Hofstede globe, a runtime fairness dashboard, a Server-Sent Events live recommendation stream and a page-aware Gemini assistant. The data layer is fully synthetic and documented; it includes a 100,000-row legacy corpus, a 60,000-row natural synthetic corpus and a live demonstration database containing 119 products, 56 users and 1002 interaction events.

Experimental results show that a LightGBM ranking baseline reaches 0.4510 NDCG@10 on the natural synthetic dataset, while a reframed binary engagement task achieves 0.9004 accuracy and 0.957 AUC. The per-innovation ablation lattice shows that CCE is neutral on engagement accuracy, CLAR reduces average top-N from 8.00 to 5.47, TDM produces a measurable recency-strength decay of 0.958, CUR detects heterogeneous uplift with a standard deviation of 0.047, and JIT-XAI adds only 4.56 ms p99 latency overhead. External validation on MovieLens-100k confirms that TDM, CUR and JIT-XAI transfer across datasets, whereas CCE and CLAR require country and behavioural signals.

The resulting system demonstrates that recommendation quality should not be reduced to a single accuracy number. Cultural transferability, cognitive gentleness, temporal memory, causal honesty and in-product explanations can be engineered together under explicit ethical constraints. Religious and political attributes are never used in model inference, an invariant enforced by tests and CI-level ethics guards.

Keywords: recommender systems, hybrid recommendation, Hofstede dimensions, cognitive

load, causal uplift, explainable AI, SHAP, ethical AI, GDPR, KVKK, FastAPI, LightGBM.

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SYMBOLS AND ABBREVIATIONS

Abbreviation / Symbol	Explanation
API	Application Programming Interface.
AUC	Area under the ROC curve; discrimination power in binary classification.
CCE	Cultural Context Embedding; Hofstede-based cultural context embedding layer.
CLAR	Cognitive Load Aware Recommender; cognitive-load-aware recommendation adaptation.
CUR	Causal Uplift Re-ranker; T-learner based causal re-ranking module.
GDPR	European Union General Data Protection Regulation.
HR@k	Hit Rate at k; the proportion of cases in which the correct item appears within the top-k recommendations.
JIT-XAI	Just-in-Time Explainability; explainability layer at the moment of decision.
KVKK	Turkish Personal Data Protection Law.
LightGBM	Gradient boosting decision tree based machine learning library.
MRR@k	Mean Reciprocal Rank at k.
NDCG@k	Normalized Discounted Cumulative Gain at k.
PBKDF2	Password-Based Key Derivation Function 2.
SHAP	Shapley Additive Explanations.
SSE	Server-Sent Events; one-way live event stream from server to browser.
TDM	Time-Decay Behavioural Memory; time-decayed behavioural memory layer.
s_{base}	Baseline hybrid recommendation score.
$\mathbf{v}_c$	Hofstede six-dimensional cultural vector of country c .
$d_H(c_a, c_b)$	Euclidean distance between two countries in Hofstede space.
L	CLAR cognitive load score.
$h_{1/2}$	TDM half-life parameter.

$\mu_1(X), \mu_0(X)$	T-learner treatment and control outcome models.
ϕ_i	SHAP/Shapley feature contribution.

1 INTRODUCTION

1.1 Context of the Study

Recommendation systems in online retail platforms are among the central decision mechanisms that determine which products users encounter, which alternatives remain visible, and which products are implicitly pushed to the margins of attention. This mechanism is therefore not merely a technical ranking problem. It directly affects information load, privacy, cultural context, the user’s right to make an informed decision, and the trust relationship established between the user and the system. A recommender that merely increases the probability of a click is no longer sufficient for an internationally defensible system. A scientifically grounded recommender must also explain why a particular product is shown, keep sensitive information outside the modelling pipeline, forget historical behaviour in a reasonable temporal manner, and ask whether the recommendation itself actually changes the user’s behaviour.

Seyr-i Öneri was developed from this motivation. The system takes a classical hybrid retail recommender architecture and extends it through five coordinated layers: cultural context, cognitive load, temporal memory, causal uplift, and just-in-time explainability. The work is therefore not presented as a single machine-learning model only. It is a complete software-engineering artefact that includes data generation, service architecture, ethical policy, automated tests, visual evidence, interactive demonstration pages, and an academically auditable experimental pipeline.

The study also has an international academic context. The English abstract of the project was accepted under Ref: AC-06060 for oral presentation within the VI. International British Congress on Interdisciplinary Scientific Research & Practices. The congress page states that the full paper may be prepared in one of the congress languages and that English abstracts are required for submission [1]. This report is therefore prepared as an extended English full paper that satisfies both conference full-text expectations and the documentation standards expected from a reproducible technical research report.

1.2 Problem Definition

Although modern recommendation systems achieve strong results at very large scales, practical deployments often contain four critical blind spots.

P1: Cultural blindness. In many systems, the user’s country is stored only as a categorical variable. For example, Türkiye, Brazil, and Japan may become unrelated integer identifiers from the model’s point of view; cultural similarities or differences that may influence consumer behaviour are not represented directly. Yet culture can influence product categories, price perception, uncertainty avoidance, and expectations about long-term utility [2, 3]. A rec-

ommendation system that treats country only as a flat label loses the possibility of explaining culturally meaningful transfer.

P2: Attention-budget blindness. Recommenders frequently present lists of the same length to every user. However, cognitive-load theory and the choice-overload literature show that more options do not always produce a better decision experience; in some contexts, additional alternatives reduce decision quality and increase fatigue [4–6]. If a user’s in-session behaviour signals intensity, hesitation, or fatigue, the system should not behave as if the user’s attention budget were unlimited.

P3: Confusing correlation with causality. A classical recommendation model usually estimates $P(Y = 1 \mid X)$. This prediction is informative about the probability that a user will choose a product, but it does not answer whether showing the product actually changes the user’s behaviour. From a business and ethics perspective, the more meaningful question is: “Would the user have selected this product even if it had not been recommended?” This question belongs to uplift modelling and causal inference rather than ordinary supervised classification [7–9].

P4: Explanations arriving too late. Explainable artificial intelligence methods such as SHAP have become widely used, but in many systems explanations remain in engineering dashboards, offline notebooks, or evaluation reports. The user, however, sees the product at the moment of decision and often receives no understandable explanation. This creates weaknesses in trust, accountability, and user agency [10–12].

1.3 Aim and Scope

The aim of this study is to design, implement, and evaluate a working hybrid retail recommendation system that addresses the four problems above within a single scientifically defensible architecture. The scope of the study includes the following components:

- a FastAPI-based web application, Jinja2 interface templates, and an SQLite data layer;
- an ethically constrained ten-feature production model contract;
- a Hofstede six-dimensional cultural embedding layer;
- cognitive-load estimation and dynamic top-N adaptation;
- time-decayed behavioural memory;
- T-learner based causal uplift re-ranking;
- SHAP-based just-in-time explainability;

- data-generation methodology, model-training scripts, ablation experiments, external-validity testing, and automated tests;
- a user interface containing a three-dimensional cultural globe, an ethics dashboard, a live recommendation stream, and a page-context-aware assistant.

The report presents the theoretical background, requirements, design decisions, data and modelling pipeline, experimental results, limitations, and future-work directions in an integrated manner. The central claim is not that every added layer automatically improves every accuracy metric. Rather, the claim is that recommendation systems should be evaluated as socio-technical decision systems in which accuracy, cognitive burden, cultural transfer, causal effect, explainability, fairness, latency, and reproducibility must be considered together.

1.4 Research Questions

The study was carried out around four main research questions. Table 1 connects each question with the corresponding evidence used in the report.

Table 1: Research questions and corresponding evidence

Code	Research Question	Measurement / Evidence
RQ1	What does adding cultural context as a continuous vector provide compared with a classical categorical country encoding?	CCE A/B experiments, cultural-distance calculations, explainability discussion, and cultural-transfer analysis.
RQ2	Can the number of recommendations be adjusted dynamically according to the user’s cognitive load?	CLAR simulation, load-band distribution, average top-N reduction, and interface-level HUD evidence.
RQ3	Does time-decayed behavioural memory and causal uplift re-ranking make the recommendation list more defensible?	TDM recency strength, CUR uplift heterogeneity, and the five-layer ablation lattice.
RQ4	Can explainability be moved to the user’s decision moment without exceeding an acceptable latency boundary?	JIT-XAI p99 latency overhead, SHAP contribution bars, and recommendation-card modal behaviour.

1.5 Original Contributions

The main contribution of this study is not the invention of a single isolated algorithm. Its contribution is the engineering integration of culture, attention, time, causality, and explainability into one working recommendation flow. Table 2 summarises the original contribution layers.

Table 2: Original contributions of the study

Contribution	Technical Content	Academic Value
CCE	Hofstede six-dimensional country vector, global-mean fallback, cultural distance, and context bonus.	Moves country information from a flat category to an interpretable continuous context.
CLAR	Cognitive-load score from five behavioural signals; four-band top-N, diversity, and confidence adaptation.	Connects the choice-overload literature to a working recommendation interface.
TDM	Ebbinghaus exponential decay, half-life parameter, familiarity penalty, and cold-start signal.	Treats user memory as temporal rather than static.
CUR	T-learner, treated/control outcome models, and uplift-blended final score.	Makes the “would have bought anyway” problem visible in ranking.
JIT-XAI	SHAP TreeExplainer, top-five contributions, counterfactual sentence, and product-card modal.	Moves explanations from the engineering panel to the user’s decision moment.
Ethical contract	Constants, tests, and CI guard excluding sensitive fields from model input.	Turns GDPR/KVKK compliance from a textual declaration into a code-level invariant.
Demo ecosystem	Three-dimensional globe, dashboard, SSE recommendation stream, and Gemini assistant.	Makes theoretical contributions observable for evaluators and users.

1.6 Organisation of the Report

The second section summarises the main concepts related to recommendation systems, cultural dimensions, cognitive load, temporal memory, causal uplift, explainable artificial intelligence, and ethics. The third section positions the study within the related literature. The fourth section explains requirements, system architecture, data model, API design, and implementation structure. The fifth section presents the material and method, including formulas for CCE, CLAR, TDM, CUR, and JIT-XAI. The sixth section reports experimental findings, baseline comparisons, ablation results, external validation, testing evidence, and fairness observations. The seventh section presents interface evidence and user-experience outputs. The eighth section discusses findings, limitations, validity threats, and ethical implications. The ninth section concludes the study and defines future work. The appendices provide the technical component inventory, selected code listings, database schema, and congress acceptance summary.

2 GENERAL CONCEPTS

2.1 Hybrid Recommendation Systems

Recommendation systems are decision-support systems that present ranked products, content, or services to users. Classical approaches are commonly grouped into collaborative filtering, content-based filtering, and hybrid methods [13–15]. Collaborative filtering learns similarity patterns from user-item interaction matrices, whereas content-based methods use item and user attributes. Hybrid systems combine these approaches in order to reduce cold-start problems, data sparsity, and overdependence on a single signal.

Seyr-i Öneri builds its hybrid structure through two model outputs and several rule-based compatibility bonuses. The store prediction model p_s , product prediction model p_p , and rule-based context bonuses are combined as follows:

$$s_{base}(u, i) = 0.55 p_s(u, i) + 0.35 p_p(u, i) + b_{cat} + b_{country} + b_{price} + b_{color}. \quad (1)$$

Here u denotes the user, i denotes the item, b_{cat} represents category fit, $b_{country}$ country fit, b_{price} price-segment fit, and b_{color} colour preference. This baseline score is subsequently extended by CCE, CLAR, TDM, CUR, and JIT-XAI. The architecture therefore begins with a recognisable hybrid recommender and then adds layers that address cultural context, attention, memory, causal effect, and explanation.

2.2 Cultural Dimensions and the Hofstede Model

The Hofstede model represents countries across six cultural dimensions: power distance (PDI), individualism (IDV), masculinity/achievement orientation (MAS), uncertainty avoidance (UAI), long-term orientation (LTO), and indulgence/restraint (IVR) [2]. The model should not be treated as a perfect or individual-level deterministic truth. Nevertheless, it offers a useful structural prior for converting country information into a richer and more interpretable representation than a single categorical label. Methodological criticisms of Hofstede’s work must also be recognised [16]. In this study, CCE is not used to reduce individuals to cultural stereotypes; it is used to make country-level average context measurable, interpretable, and available for transparent analysis.

2.3 Cognitive Load and Choice Overload

Cognitive-load theory argues that human working memory is limited and that learning or decision quality may decline when too much information is processed simultaneously [4]. Iyengar and Lepper’s choice-overload experiment similarly showed that more options do

not necessarily produce better user experience [5]. This issue is particularly important in recommender systems: a system may be capable of showing the best twenty products, while the user may only be able to evaluate three products carefully at that moment.

CLAR therefore changes the role of the recommendation system. Instead of acting only as a score sorter, it attempts to estimate the user’s current attention budget and adapt list length, confidence threshold, and diversity accordingly. This does not replace ranking accuracy; it complements it with a user-centred constraint.

2.4 Temporal Memory

User preferences are not stationary. Interacting with a product yesterday and interacting with it six months ago should not have the same meaning. Time-aware recommenders accept that the representational value of past events decreases over time [17, 18]. The TDM module in Seyr-i Öneri operationalises this idea by applying the Ebbinghaus forgetting curve at recommendation-score level. The purpose is not to erase history, but to stop the system from treating stale behaviour as if it were equally current.

2.5 Causal Uplift

Causal uplift attempts to estimate the incremental effect of an action on a particular user. In a recommendation context, the action is exposing the user to the product. A classical model estimates $P(Y = 1 \mid X)$, whereas an uplift model asks for the difference:

$$\tau(X) = \mathbb{E}[Y \mid do(T = 1), X] - \mathbb{E}[Y \mid do(T = 0), X]. \quad (2)$$

If this difference is positive, the user may be considered *persuadable*; if it is negative, the user may behave as a *sleeping dog*; if it is close to zero, the recommendation has weak incremental effect. The T-learner approach estimates this difference by training separate treatment and control models [7]. CUR introduces this causal question into the ranking stage.

2.6 Explainable Artificial Intelligence

Explainable artificial intelligence aims to make model outputs understandable to users, developers, auditors, and decision-makers. SHAP uses Shapley values from cooperative game theory to calculate each feature’s contribution to a prediction [10]. For tree-based models, TreeExplainer can perform this calculation efficiently. Seyr-i Öneri does not produce explanations only as offline reports. It exposes them next to recommendation cards so that the user can see why a product appears at the moment of decision.

2.7 Ethics, Privacy, and Fairness

Recommendation systems are not ethically neutral because they shape user behaviour and allocate attention. Including sensitive attributes in model input can generate discrimination, bias, and privacy violations [19, 20]. Seyr-i Öneri therefore keeps sensitive fields outside the model. Religious and political orientation fields may exist as optional form or database fields, but they are excluded from the model feature set. This distinction is one of the most important ethical design choices in the project: a field can be present for user expression or profile completeness without becoming a training signal.

3 LITERATURE REVIEW

3.1 Modern Recommendation Systems

Ricci et al. provide a comprehensive framework that describes recommendation systems in terms of users, items, context, and feedback [13]. Koren et al. explain the foundations of collaborative filtering and matrix factorisation [14]. Deep-learning-based recommenders have been influential, especially in large-scale candidate generation and ranking problems such as YouTube recommendations [21]. However, Dacrema et al. showed that new deep recommendation methods do not always outperform strong baselines and that reproducibility is critical [22]. For this reason, Seyr-i Öneri gives importance not only to model claims, but also to baseline comparisons, ablation experiments, and reproducible artefacts.

3.2 Culture- and Personality-Based Recommendation

Culture- and personality-aware recommenders have been studied in human-computer interaction and user-modelling research [23, 24]. Many of these studies, however, do not provide an end-to-end, production-proximal, open, and visually inspectable retail recommendation application. Seyr-i Öneri aims to fill this gap by connecting Hofstede vectors to a working FastAPI application, the scoring layer, the data pipeline, and the user interface.

3.3 Cognitive Load and User Experience

Bollen et al. experimentally examined the effect of recommendation-list length on user experience and decision difficulty [6]. Knijnenburg et al. argued that recommender systems should be evaluated not only by accuracy, but also by perceived user effort and experience quality [25]. CLAR translates this literature into runtime system behaviour: when load is high, the system prefers fewer but more confident recommendations.

3.4 Temporal Recommendation and Forgetting Mechanisms

Ding and Li proposed time-weighted collaborative filtering, in which historical interactions are weighted by recency [17]. Vinagre et al. classified forgetting mechanisms in streaming recommender settings [18]. The TDM module follows this line of work but adds an interpretable half-life parameter that can be explained to evaluators, developers, and users.

3.5 Causal Inference and Uplift Modelling

Kunzel et al. formalised T-, S-, and X-learner meta-algorithms for heterogeneous treatment-effect estimation [7]. Athey and Imbens studied heterogeneous effect decomposition using causal trees [8]. Diemert et al. provided a large-scale benchmark for uplift modelling [9]. CUR translates these ideas into a recommendation re-ranking layer and explicitly separates high probability from high incremental effect.

3.6 Explainable Recommendation Systems

Lundberg and Lee introduced the SHAP framework, while Zhang and Chen provided a broad survey of explainable recommendation systems [10, 12]. Goodman and Flaxman discuss the legal and ethical dimensions of the right to explanation in automated decision-making systems [11]. The JIT-XAI module in Seyr-i Öneri moves the explanation from offline analysis into the product-card experience.

3.7 Design Principles Derived from the Literature

The literature review leads to five design principles for this study:

1. Recommendation systems should not be defended with a single metric; accuracy, ranking quality, diversity, latency, fairness, and explainability should be reported together.
2. Country and culture information should be used as interpretable context, not as deterministic judgement about individuals.
3. The user interface should make model decisions visible rather than hiding them behind a black-box ranking list.
4. Causal claims require real treatment logs; when logs are unavailable, proxy use must be declared as a limitation.
5. Ethical decisions must be protected not only in the written report, but also through constants, tests, and CI controls in the codebase.

4 RELATED WORK AND POSITIONING

This section summarises how the study is positioned within the literature on recommendation systems, cultural-context modelling, cognitive load, temporal memory, causal uplift, and explainable artificial intelligence. The detailed theoretical discussion was provided in the previous section; here, the focus is on how Seyr-i Öneri targets the gap between separate research threads and an end-to-end working system.

The core difference of Seyr-i Öneri cannot be reduced to a single algorithmic contribution. The system presents CCE, CLAR, TDM, CUR, and JIT-XAI within one FastAPI application, together with an ethical feature contract, automated tests, reproducible artefacts, and an inspectable user interface. In this respect, the work is not merely an experiment that reports model performance; it is an academically auditable software-engineering product.

In the literature, culture, attention, time, causality, and explainability are often investigated separately. This study’s contribution is to connect these five dimensions to the same recommendation flow and to support each one with either a measurable metric or an observable interface behaviour. The design therefore makes the scientific claims visible at three levels: the mathematical formulation, the experimental result, and the product experience.

The positioning of the work is deliberately conservative in its empirical claims. CCE is not presented as a universal accuracy booster; CLAR is not evaluated as if cognitive load were the same as classification accuracy; CUR is not claimed to prove causal effect without real randomised exposure logs; and JIT-XAI is not treated as a complete solution to transparency. Instead, the system is positioned as a reproducible and expandable reference implementation showing how these concerns can be engineered together under explicit ethical constraints.

5 SYSTEM REQUIREMENTS AND DESIGN

5.1 Requirement Analysis

The system requirements were grouped into three categories: functional requirements, non-functional requirements, and ethical requirements. This separation is important because the project is not only a prediction pipeline; it is an interactive software system with explicit scientific, usability, reproducibility, and ethical obligations.

Table 3: Functional requirements

Code	Requirement	Realisation
FR-01	Users must be able to register and log in.	Session-based authentication and PBKDF2 password hashing.
FR-02	A recommendation feature frame must be generated from the user profile.	Feature-frame generation inside the recommendation service.
FR-03	The catalogue must be listed, filtered, and opened at product-detail level.	Catalogue and product-detail routes.
FR-04	User interactions must be recorded.	Event repository and interaction records.
FR-05	The system must not stop working when model artefacts are unavailable.	Popularity- and preference-based fall-back ranker.
FR-06	CCE, CLAR, TDM, CUR, and JIT-XAI must run as separate services.	Independent service modules.
FR-07	Live explanation and live recommendation flow must be shown in the interface.	XAI modal and SSE stream.
FR-08	Ethical and fairness status must be monitorable.	Dashboard and fairness snapshot API.

Table 4: Non-functional and ethical requirements

Code	Requirement	Implementation Counterpart
NFR-01	The system must provide a fast local demonstration.	Batch launcher, SQLite, and Jinja2 templates.
NFR-02	Model artefacts must be reproducible.	Git SHA, dataset hash, and seed 42.
NFR-03	Explanation generation must be fast enough not to harm user experience.	JIT-XAI p99 overhead of 4.56 ms.
NFR-04	Sensitive fields must not enter model input.	Ten-feature contract, automated tests, and CI guard.
NFR-05	The code must be testable.	Repository/service separation and 80 automated tests.
NFR-06	Visual evidence must be available for academic presentation.	Globe, dashboard, innovation demos, and screenshots.

5.2 High-Level Architecture

The system follows a layered architecture. The presentation layer is built with Jinja2 templates and vanilla JavaScript. The controller layer manages HTTP requests through the FastAPI router. The repository layer communicates with SQLite tables. The service layer contains recommendation, cultural embedding, cognitive load, temporal memory, causal uplift, explainability, and chatbot logic. This separation keeps the scientific modules testable while allowing the user interface to expose their outputs.

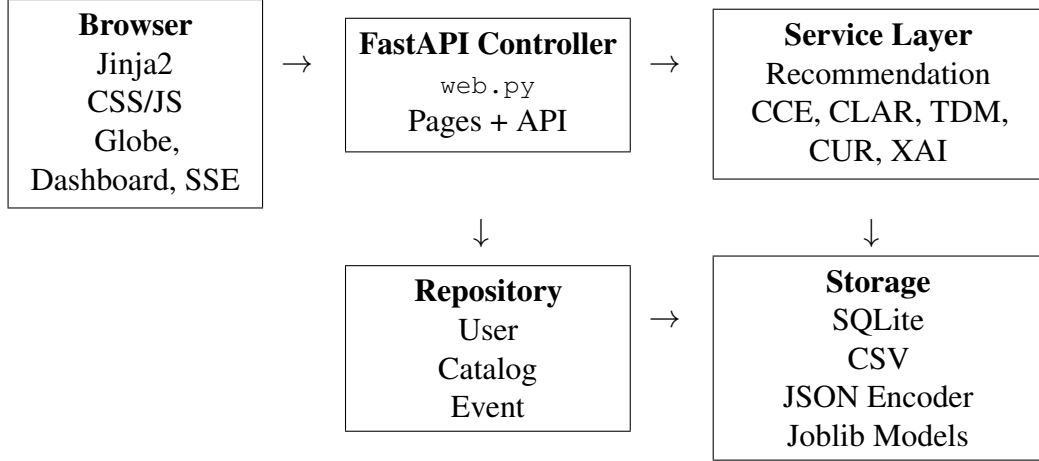


Figure 1: High-level system architecture of Seyr-i Öneri

5.3 Project Structure

The project is organised so that web controllers, repositories, scientific services, training scripts, tests, and artefacts can be inspected separately. This structure supports both engineering maintainability and academic reproducibility.

```

1 app/
2   main.py                # FastAPI app and lifespan
3   config.py              # path and secret configuration
4   constants.py           # ethical feature contract and feature flags
5   db.py                  # SQLite schema
6   controllers/web.py     # pages and API endpoints
7   repositories/          # user, catalog, event repository
8   services/
9     recommendation_service.py
10    cultural_embedding.py
11    cognitive_load.py
12    temporal_memory.py
13    causal_uplift.py
14    explainability.py
15    chatbot_service.py
16   templates/             # Jinja2 UI
17   static/                 # CSS and JavaScript
18   scripts/               # training, data generation, ablation, reports
19   tests/                 # unit and integration tests
20   artifacts/             # model and scientific result artefacts
21   manuscript/            # congress and full-text files

```

Code Listing 1: Simplified project directory tree

5.4 Data Model

The SQLite database is built on three core tables: user profiles, product catalogue, and user interactions. This minimal schema is sufficient for the demonstration flow, controlled experiments, and reproducible recommendation examples.

Table 5: SQLite table summary

Table	Current Rows	Responsibility
users	56	User profile fields, authentication information, ethically optional sensitive fields, and preference signals.
products	119	Product name, category, colour, price segment, country, store information, popularity, and average satisfaction.
user_events	1002	Product views, dwell time, satisfaction score, and time-stamped interaction records.

5.5 Ethical Feature Contract

One of the most important design decisions in Seyr-i Öneri is that the production model feature set is fixed and ethically constrained. Religious and political orientation fields may remain in the database as optional expression fields, but they cannot be included in the model feature set. This design makes the ethical boundary technically enforceable.

Table 6: Ten ethical features used in the production model

Feature	Meaning
enc_country	Encoded country field of the user.
enc_city	Encoded city field of the user.
enc_county	Encoded county/region field of the user.
enc_gender	Encoded gender field of the user.
z_age	Z-score normalised age.
z_spent_on_product_page	Z-score normalised time spent on the product page.
enc_the_category_of_product	Preferred product category.
enc_color_of_the_product	Preferred product colour.
is_product_expensive_int	Price-segment preference.
enc_shopping_frequency	Shopping-frequency signal.

When CCE is enabled, only six additional columns derived from country-level Hofstede statistics may be used: `cce_pdi`, `cce_idv`, `cce_mas`, `cce_uai`, `cce_lto`, and `cce_ivr`. These columns are not individual sensitive data; they are country-level cultural-context variables. This distinction is essential for the ethical defensibility of the model.

5.6 API and Interface Design

The system is not composed only of backend functions. Each scientific contribution is made visible to the user or evaluator through a dedicated interface surface.

Table 7: Main pages and API endpoints

Route	Function	Related Contribution
/recommendations	Personal recommendation list, JIT-XAI modal, and SSE flow.	All innovation layers.
/innovations/cce	Country selection and Hofstede-distance demonstration.	CCE.
/innovations/clar	Behaviour-signal sliders and cognitive-load band.	CLAR.
/innovations/tdm	Half-life slider and exponential decay curve.	TDM.
/innovations/cur	Uplift-distribution simulator.	CUR.
/innovations/xai	Feature sliders and SHAP bars.	JIT-XAI.
/globe	Three-dimensional cultural globe, country density, and cultural neighbours.	CCE visualisation.
/dashboard	Fairness snapshot, cognitive-load status, and ethical badge.	Ethics and CLAR.
/api/recommendations/stream	Sends recommendations as a live event stream.	SSE demonstration.
/api/explain/key	Generates an immediate explanation for a product.	JIT-XAI.

6 MATERIAL AND METHOD

6.1 Data Layers

No real user data were used in the study. The data were organised into three conceptual layers so that legacy experiments, natural synthetic evaluation, and live interface demonstration could be discussed separately.

Table 8: Data layers and purposes of use

Layer	File / Source	Size	Use
L0 Legacy	person.csv	100,000 rows, 16 columns	Initial training and classical baseline comparisons.
L1 Legacy ML	person_normalized_ml.csv	100,000 rows, 31 columns	Model training with encoded features and target variables.
L2 Natural	data/natural/person.csv	60,000 rows, 16 columns	Scientific experiments with a more realistic conditional-probability chain.
L2 Natural ML	data/natural/person_normalized_ml.csv	60,000 rows, 31 columns	Scientific baseline, binary engagement, and ablation experiments.
Demo DB	seyr_i_oneri.db	119 products, 56 users, 1002 interactions	Live interface and congress demonstration.

In the natural synthetic data-generation process, country, age, gender, shopping frequency, category, price segment, page duration, and satisfaction score were generated through a conditional probability chain. Product popularity was based on a Pareto distribution, dwell time on a Gamma distribution, and satisfaction on a Beta-based Likert transformation. These choices are consistent with retail long-tail behaviour and web dwell-time literature [26–28].

6.2 Cultural Context Embedding

Each country c is transformed into a six-dimensional vector composed of Hofstede scores:

$$\mathbf{h}_c = [PDI_c, IDV_c, MAS_c, UAI_c, LTO_c, IVR_c]. \quad (3)$$

The vector is normalised to the interval $[0, 1]$:

$$\mathbf{v}_c = \frac{\mathbf{h}_c}{100}. \quad (4)$$

The cultural distance between two countries is computed as follows:

$$d_H(c_a, c_b) = \|\mathbf{v}_{c_a} - \mathbf{v}_{c_b}\|_2. \quad (5)$$

At scoring time, if the cultural distance between the user’s country and the product’s country is below 0.30, the product receives a small context bonus:

$$b_{culture}(u, i) = \begin{cases} 0.02, & d_H(country_u, country_i) < 0.30, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

The numerical value of the bonus is intentionally small. The goal is not to override the main model, but to make culturally close alternatives slightly more visible while keeping the ranking primarily driven by behavioural and product signals.

6.3 Cognitive Load Estimation

CLAR uses five behavioural signals observed during a session:

$$X_{clar} = [\sigma_d, v_c, h, t, b], \quad (7)$$

where σ_d denotes dwell-time variance, v_c click velocity, h hover count, t scroll-thrash ratio, and b short-bounce ratio. Each signal is normalised and passed through a sigmoid compressor:

$$\sigma(z) = \frac{1}{1 + e^{-4z}}. \quad (8)$$

The cognitive-load score is calculated as follows:

$$L = 0.25s_{dwell} + 0.25s_{click} + 0.15s_{hover} + 0.20s_{scroll} + 0.15s_{bounce}. \quad (9)$$

Table 9: CLAR adaptation table

Band	L Range	Top-N	Diversity Boost	Min. Confidence
Low	$[0, 0.25)$	12	1.0	0.10
Moderate	$[0.25, 0.50)$	8	1.1	0.15
High	$[0.50, 0.75)$	5	1.3	0.25
Overload	$[0.75, 1]$	3	1.5	0.35

This table translates cognitive-load estimation into visible recommendation behaviour. In low-load sessions, the system can present a longer list. In overload sessions, it intentionally shortens the list and raises the confidence threshold.

6.4 Time-Decayed Behavioural Memory

For each event, TDM takes the age in days as Δt and computes the following weight:

$$w(\Delta t) = \exp\left(-\frac{\ln 2}{h_{1/2}}\Delta t\right). \quad (10)$$

The default half-life is $h_{1/2} = 14$ days. The familiarity value of a product is then calculated as:

$$fam_i = \frac{\sum_{e \in E_i} w(\Delta t_e)}{\sum_{e \in E} w(\Delta t_e)}. \quad (11)$$

This value is subtracted from the hybrid score as a small diversity-oriented penalty:

$$s_{tdm}(u, i) = s_{base}(u, i) - 0.05 fam_i. \quad (12)$$

The penalty prevents the system from repeatedly recommending items that are already familiar while still preserving recent positive behaviour.

6.5 Causal Uplift Re-Ranking

CUR uses a T-learner. Two separate regressors are trained for treated and untreated samples:

$$\mu_1(X) = \mathbb{E}[Y \mid T = 1, X], \quad \mu_0(X) = \mathbb{E}[Y \mid T = 0, X]. \quad (13)$$

The uplift score is:

$$\tau(X) = \mu_1(X) - \mu_0(X). \quad (14)$$

The final blended score is:

$$\hat{s}_i = s_i + \lambda \tau_i, \quad \lambda = 0.15. \quad (15)$$

This approach allows the recommendation system to promote not only high-probability products, but also products for which recommendation exposure is more likely to create incremental behavioural change. Because the available data do not contain real randomised exposure logs, the current implementation uses a proxy treatment signal and reports this limitation explicitly.

6.6 Just-in-Time Explainability

SHAP contributions are based on Shapley values:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)]. \quad (16)$$

For each recommendation, Seyr-i Öneri presents the five features with the largest absolute contribution. Positive contributions are shown with green bars and negative contributions with red bars. The system also provides a short counterfactual sentence to the user: “If this feature changed in the opposite direction, the product’s position would probably decrease/increase.” The explanation is therefore not restricted to a technical report; it becomes part of the recommendation card.

6.7 Evaluation Metrics

The study does not rely on a single success metric. Ranking quality, binary discrimination, list properties, latency, and statistical confidence are reported together.

Table 10: Evaluation metrics

Metric	Definition	Use
NDCG@10	Discounted gain according to the position of the correct item.	Ranking model quality.
MRR@10	Mean reciprocal rank of the first correct item.	Speed of reaching the first correct recommendation.
HR@10	Proportion of cases where the correct item appears in the top 10.	Top-k capture performance.
Accuracy	Ratio of correct classifications.	Binary engagement and category prediction.
AUC	Area under the ROC curve.	Binary discrimination power.
List Diversity	Within-list category diversity.	Measurement of monotone recommendation risk.
Catalog Coverage	Proportion of catalogue covered by recommendations.	Long-tail health.
p99 Latency	Latency below which 99 percent of requests fall.	User experience and JIT-XAI cost.
Wilcoxon p-value	Paired statistical significance test.	Reliability of A/B differences such as CCE.

7 IMPLEMENTATION AND REALISATION

7.1 FastAPI Startup Flow

The application starts with the modern FastAPI lifespan approach. At startup, the database schema is created and the catalogue is seeded from CSV files if the catalogue is empty.

This startup pattern makes the local demonstration repeatable and avoids manual database preparation.

```
1 @asynccontextmanager
2 async def lifespan(app: FastAPI) -> AsyncIterator[None]:
3     initialize_database()
4     catalog_repository.seed_if_empty()
5     yield
6
7 app = FastAPI(
8     title="Seyr-i Oneri",
9     version="0.2.0",
10    lifespan=lifespan,
11 )
12 app.add_middleware(SessionMiddleware, secret_key=SECRET_KEY, same_site="lax")
13 app.mount("/static", StaticFiles(directory=str(STATIC_DIR)), name="static")
14 app.include_router(router)
```

Code Listing 2: FastAPI lifespan startup flow

7.2 Recommendation-Service Orchestration

The recommendation service coordinates profile features, behavioural statistics, fallback ranking, model inference, cultural bonus, temporal penalty, causal uplift, and explanation generation. The following listing summarises the operational flow.

```
1 def recommend_for_user(user, top_n=None, return_context=False):
2     stats = event_repository.get_user_stats(user["id"])
3
4     if ENABLE_COGNITIVE_LOAD_ADAPTATION:
5         load = load_estimator.estimate(signals_from_event_stats(stats))
6         effective_top_n = top_n or load.adaptation.top_n
7
8     if ENABLE_TEMPORAL_DECAY:
9         events = event_repository.recent_events(user["id"], limit=50)
10        temporal = TemporalMemory(half_life_days=14).aggregate(events)
11
12    if not models_ready:
13        return fallback_recommendations(user)
14
15    features = build_feature_frame(user, stats)
16    product_p = product_model.predict_proba(features)
17    store_p = store_model.predict_proba(features)
18
19    for product in catalog:
20        score = 0.55 * store_p + 0.35 * product_p
21        score += category_bonus + country_bonus + price_bonus + color_bonus
22        score += cultural_bonus
23        score -= temporal_familiarity_penalty
24
25    if causal_uplift_is_fitted:
26        candidates = uplift_reranker.rerank(candidates, features)
27
28    explanations = shap_explainer.explain(top_results)
29    return top_results, context
```

Code Listing 3: Simplified recommendation flow pseudocode

7.3 User Security and Password Storage

Passwords are not stored in plain text. The system generates password hashes with PBKDF2-HMAC-SHA256 and a random salt. During verification, a constant-time comparison is performed with `hmac.compare_digest`. This design avoids the plain-text password mistake that is common in simple demonstration applications and makes the authentication layer defensible for an academic software prototype.

7.4 Chatbot Layer

The additional Seyri assistant runs through the Gemini 2.5 Flash API. The assistant receives the page title, visible-text summary, product cards, and quick actions as context. If the API key is unavailable, the system informs the user that the assistant is disabled instead of failing the main application. This design prevents external service interruption from breaking the demonstration.

7.5 Visual Interface and Demo Mode

The interface was developed with a dark-first design system. The home page, innovation showcase, three-dimensional cultural globe, ethics dashboard, and recommendation screen share the same visual language. Demo Mode automatically navigates through pages during a congress presentation and highlights where each innovation is triggered. The visual layer is therefore not decorative only; it is part of the evidence strategy of the study.

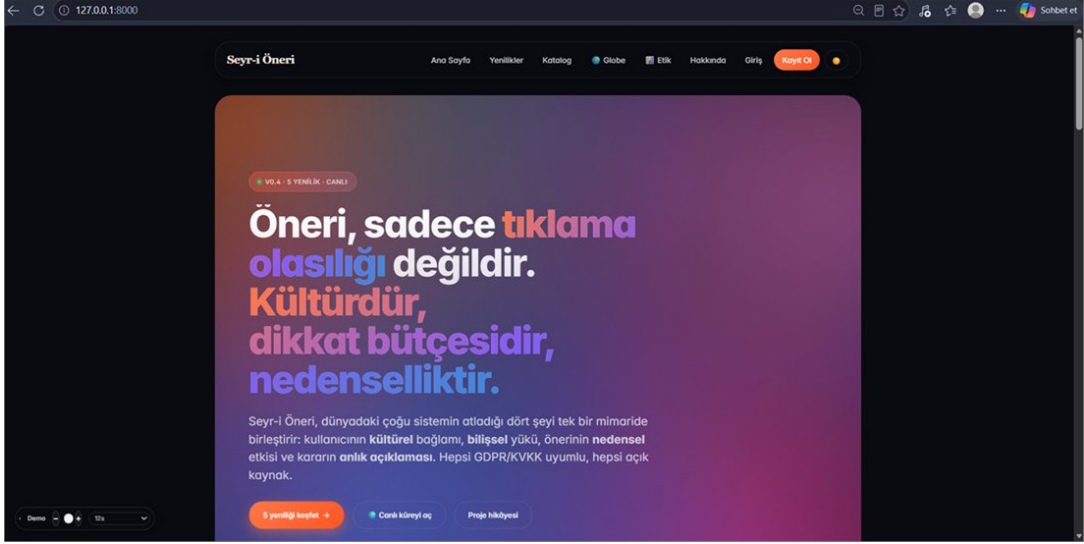


Figure 2: Seyr-i Öneri home page: a dark-first interface combining the problem statement, contributions, and demo flow.

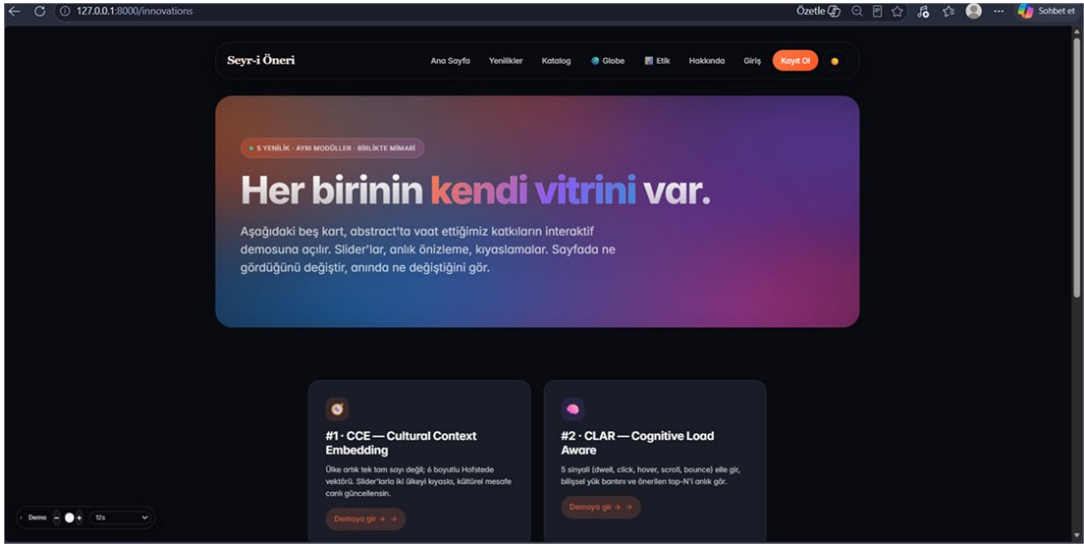


Figure 3: Five-innovation showcase: CCE, CLAR, TDM, CUR, and JIT-XAI are connected to separate demonstration pages.

8 RESEARCH RESULTS AND FINDINGS

8.1 Model Baseline Results

Within the scientific training framework conducted on the natural synthetic dataset, Random sanity, Popularity, KNN, SVD-logistic, Logistic Regression, Random Forest, RF+CCE, MLP, Gaussian Naive Bayes, and LightGBM models were compared. The purpose of this comparison was to verify that the pipeline learns real signal beyond random or popularity-only behaviour, and to position the proposed innovation layers against strong classical baselines.

Table 11: Baseline ranking results on the natural synthetic dataset

Model	Features	NDCG@10	MRR@10	HR@10	Fit Time (s)
Random sanity	10	0.0415	0.027	0.090	0.0
Popularity	10	0.0727	0.048	0.156	0.0
KNN	10	0.2181	0.157	0.420	0.3
SVD + Logistic	10	0.1808	0.124	0.370	11.1
Logistic Regression	10	0.3424	0.235	0.699	25.0
Random Forest	10	0.4487	0.309	0.909	86.9
Random Forest + CCE	16	0.4454	0.308	0.898	77.8
MLP	10	0.4503	0.311	0.909	32.1
Gaussian NB	10	0.4505	0.310	0.913	0.2
LightGBM	10	0.4510	0.310	0.917	276.0

The result is important for two reasons. First, random and popularity sanity baselines remain clearly behind the learning-based models; therefore, the model pipeline learns meaningful signal. Second, CCE does not automatically improve the category/ranking task. This negative or neutral observation was not hidden. On the contrary, it is reported as one of the scientific honesty points of the study: cultural embedding should be used when the task and the signal support it, not as an automatic improvement switch.

8.2 CCE A/B Findings

The BASE and BASE+CCE comparison conducted on the older 100,000-row dataset shows that product prediction and store prediction behave differently.

Table 12: CCE A/B comparison: product and store prediction

Task	Configuration	Features	Accuracy	NDCG@10	MRR@10	HR@10
Product	BASE	10	0.1873	0.5815	0.4459	1.0000
Product	BASE+CCE	16	0.1920	0.5842	0.4499	0.9990
Store	BASE	10	0.1110	0.4582	0.3005	0.9903
Store	BASE+CCE	16	0.1033	0.4422	0.2918	0.9465

CCE provides a small but positive contribution to product prediction. In store prediction, however, CCE decreases performance. This suggests that store choice may be more strongly associated with local signals such as delivery distance, store coverage, and campaign availability than with country-level cultural context. Therefore, CCE should not be added blindly to every prediction task. It should be activated when the target task can plausibly benefit from cultural similarity, cultural transfer, or cold-start context.

8.3 Binary Engagement Reframing

Category prediction can sometimes be dominated by demographic priors. For this reason, the study also evaluates a binary engagement/satisfaction task that better represents user-product fit. The target variable is defined as follows:

$$y_{engaged} = \begin{cases} 1, & dwell \geq p_{70}(dwell) \wedge satisfaction \geq 4, \\ 0, & \text{otherwise.} \end{cases} \quad (17)$$

Table 13: Binary engagement / satisfaction results

Configuration	Features	Accuracy	AUC	F1
BASE	10	0.9004	0.957	0.770
BASE+CCE	16	0.8999	0.957	0.769
BASE+CCE+Interaction	24	0.8996	0.957	0.769

The result shows that a well-tuned ten-feature LightGBM baseline is already strong for the binary engagement task. CCE produces neither meaningful improvement nor serious degradation in this setup. Therefore, the academic value of CCE should not be evaluated only through accuracy. It must also be assessed through interpretability, cultural transfer, fairness, cold-start reasoning, and user-facing explanation.

8.4 Ablation Lattice for the Five Innovations

Because each innovation serves a different purpose, the five contributions are not forced into a single metric. The ablation lattice evaluates every contribution with its own operational measurement.

Table 14: Ablation lattice for the five innovation layers

Innovation	Metric	Baseline	With Innovation	Interpretation
CCE	Engagement accuracy	0.8916	0.8914	Neutral; $p = 0.875$.
CLAR	Average top-N	8.00	5.47	The list becomes shorter under cognitive load.
TDM	Recency strength	1.000	0.958	Exponential decay is active.
CUR	Uplift std	0.000	0.047	Heterogeneous treatment effect is captured.
JIT-XAI	SHAP p99 overhead	2.12 ms	6.68 ms	+4.56 ms; below the 50 ms threshold.

This table is one of the most important defence points of the study. The system is not merely a model that tries to increase accuracy. It is a software architecture that makes user experience, explanation, temporal sensitivity, and causal-effect distinction measurable.

8.5 MovieLens-100k External Validity Test

The external validity test conducted on the MovieLens-100k dataset shows which innovations can transfer independently of the original dataset and which ones require particular signals.

Table 15: MovieLens-100k external validity results

Component	Value	Interpretation
BASE LightGBM Accuracy	0.6879	Reasonable implicit-feedback baseline range for ML-100k.
BASE LightGBM AUC	0.7456	Binary engagement discrimination power.
TDM $h_{1/2} = 14$	0.190	Recency is low because the dataset contains older timestamps.
TDM $h_{1/2} = 180$	1.000	The half-life parameter works in the expected direction.
CUR uplift mean	+0.175	Popularity treatment effect is strong.
CUR uplift std	0.104	Heterogeneity is higher than in the synthetic dataset.
JIT-XAI p99 overhead	+8.87 ms	Higher than the synthetic setup but below the 50 ms threshold.

MovieLens does not contain country information or in-session behavioural signals. Therefore, CCE and CLAR cannot be applied directly. This is not a failure; it is a clear definition of the boundary of validity. TDM, CUR, and JIT-XAI, however, can be transferred through timestamp, treatment proxy, and model-explanation mechanisms.

8.6 Test and Coverage Results

The codebase was verified with 80 automated tests. According to the coverage report, 1022 of 1362 statements were covered, corresponding to approximately 0.7504 line coverage. This level of testing is important because the system contains not only model code, but also security, ethics, repository, service, metric, and interface-supporting logic.

Table 16: Module-level summary of test coverage

Test Group	Risk Covered
Auth and Config	Password hashing, secret requirement, and secure session behaviour.
Repository and DB	SQLite schema, user registration, event aggregation, and unique email constraint.
Metrics	NDCG, MRR, HR, diversity, and coverage calculations.
CCE	Country resolution, fallback, vector construction, distance, and neighbours.
CLAR	Cognitive-load components, band mapping, and adaptation.
TDM	Half-life, event weight, cold-start, and familiarity.
CUR	T-learner fitting, insufficient-data fallback, and uplift prediction.
JIT-XAI	Fallback explanation, contribution ordering, and payload structure.
RecommendationService	Ten-feature contract, fallback path, and hybrid bonuses.
Ethics Guard	Verification that sensitive fields are absent from the model feature set.

8.7 Fairness Snapshot

Fairness outputs summarise the distribution of accuracy, HR@10, and F1 across country, gender, and age-band groups. In product prediction, HR@10 disparity is 0.0 across all groups, meaning that the top-10 hit rate appears equal at group level. Accuracy and F1 differences are more limited but still require monitoring. This result does not claim that the system has fully solved fairness. Rather, it shows that the system has the infrastructure needed to move fairness indicators into a runtime dashboard.

8.8 Alignment with the Accepted Congress Paper

The acceptance record lists the study title as “Design and Implementation of a Culturally Grounded, Cognitively Aware, and Causally Honest Hybrid Retail Recommendation System: Seyr-i Öneri” and states that the paper was accepted within the VI. International British Congress for oral presentation and proceedings inclusion. The congress information page provides the full-paper framework and submission context [1]. This report expands the accepted abstract into a full English manuscript and technical report with method, findings, discussion, references, and appendices.

9 INTERFACE, VISUAL EVIDENCE, AND USER EXPERIENCE

9.1 Innovation Showcase

The innovation showcase ensures that the five scientific contributions are not described only in text, but are experienced through working demonstration pages. This design decision makes

the project more convincing for evaluators and congress participants because each scientific layer can be observed directly.

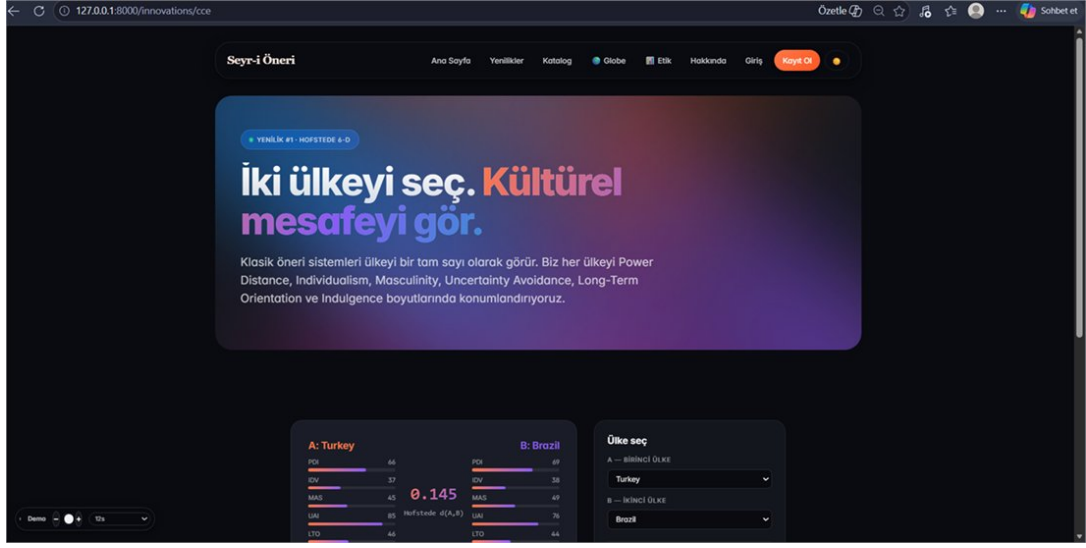


Figure 4: CCE demonstration page: Hofstede dimensions and cultural distance between two countries are visualised.

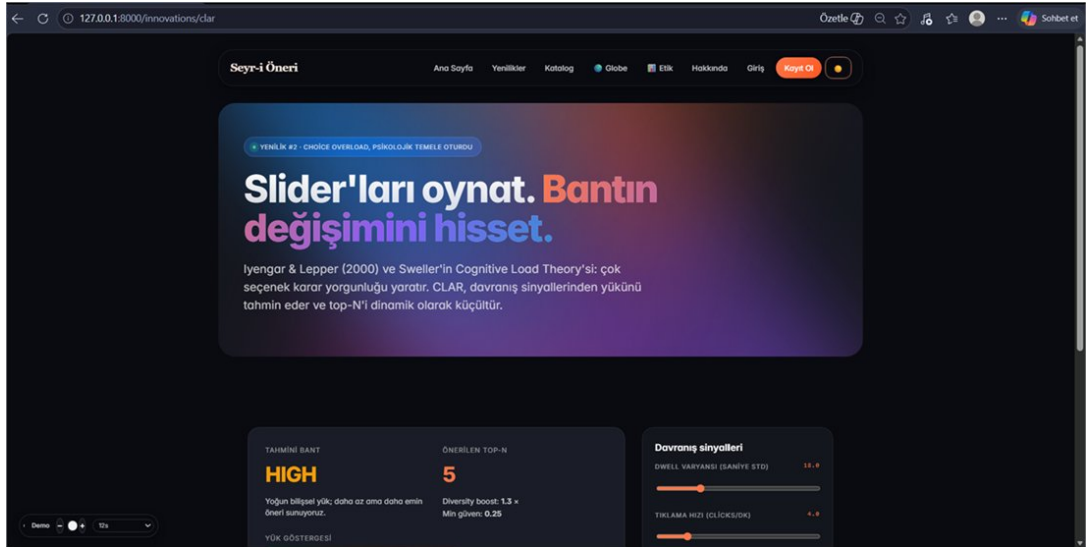


Figure 5: CLAR demonstration page: five behavioural signals are transformed into a cognitive-load band.

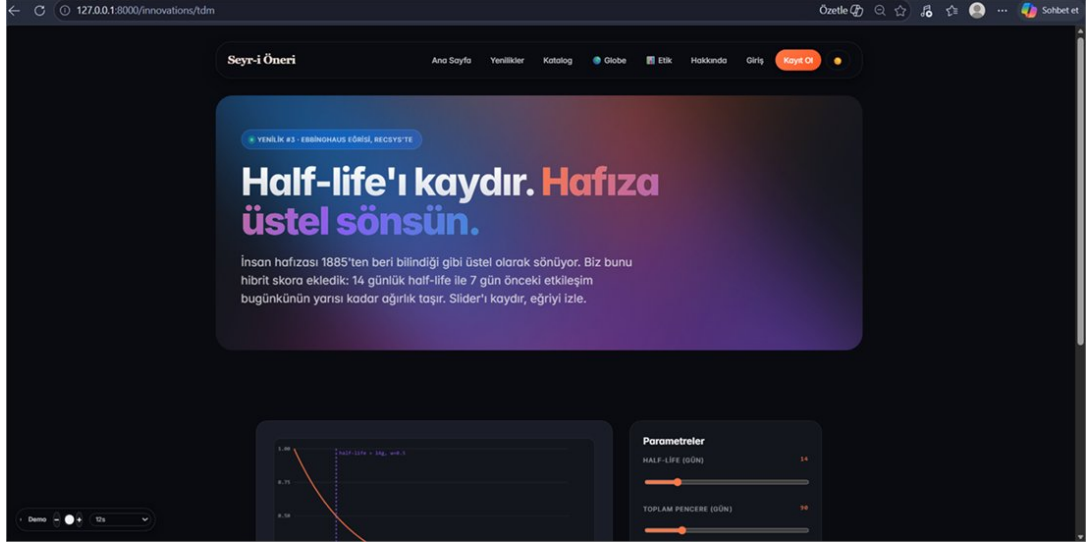


Figure 6: TDM demonstration page: the exponential decay curve is displayed with a half-life parameter.



Figure 7: CUR demonstration page: uplift distribution and persuadable/sleeping-dog distinction are simulated.

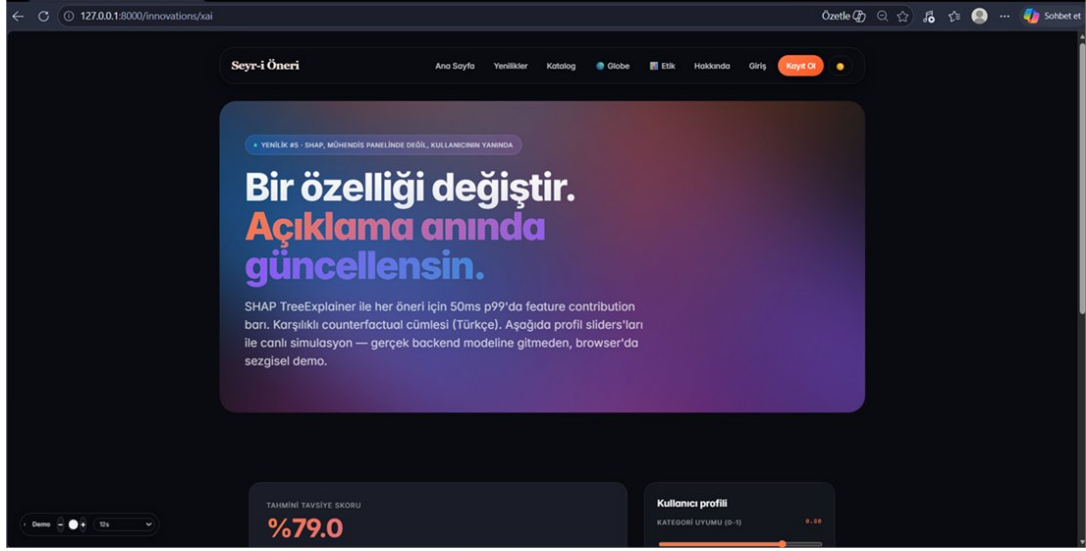


Figure 8: JIT-XAI demonstration page: feature contributions are presented through SHAP-like bars.

9.2 Three-Dimensional Cultural Globe

The globe page presents country densities and Hofstede vectors on a three-dimensional world. When the user selects a country, the cultural dimensions and nearest cultural neighbours are displayed. This page visually demonstrates that CCE is not only a feature-engineering trick; it is an interpretable cultural-context layer.

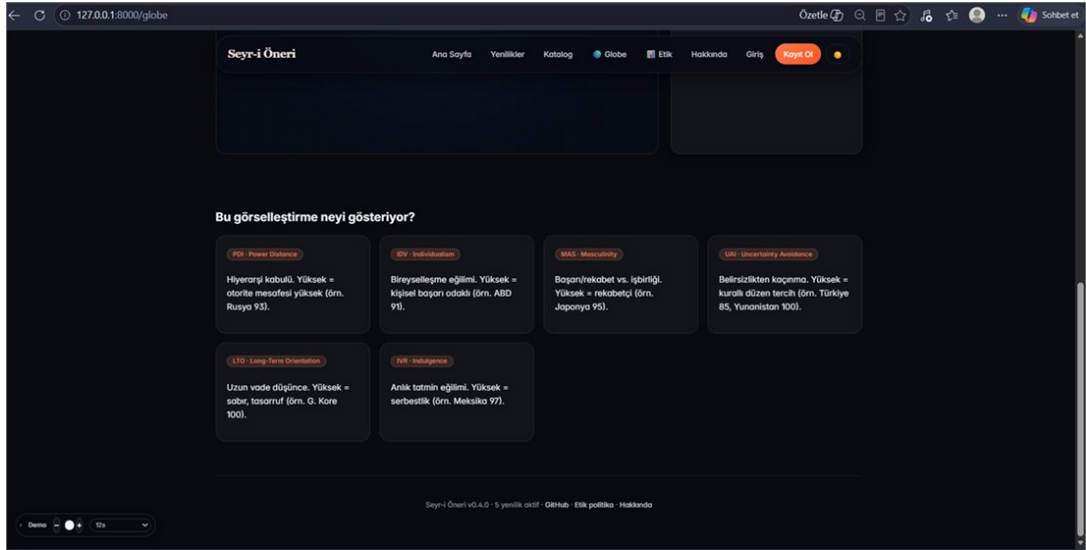


Figure 9: Three-dimensional Hofstede cultural globe: country density, cultural vector, and neighbourhood information.

9.3 Ethics and Fairness Dashboard

The dashboard page makes the system's ethical decisions visible. It emphasises that sensitive attributes are excluded from the model; demographic distributions and parity differences are

monitored through charts. In this way, ethical policy is not only documented in the report but is also given a place in the interface.

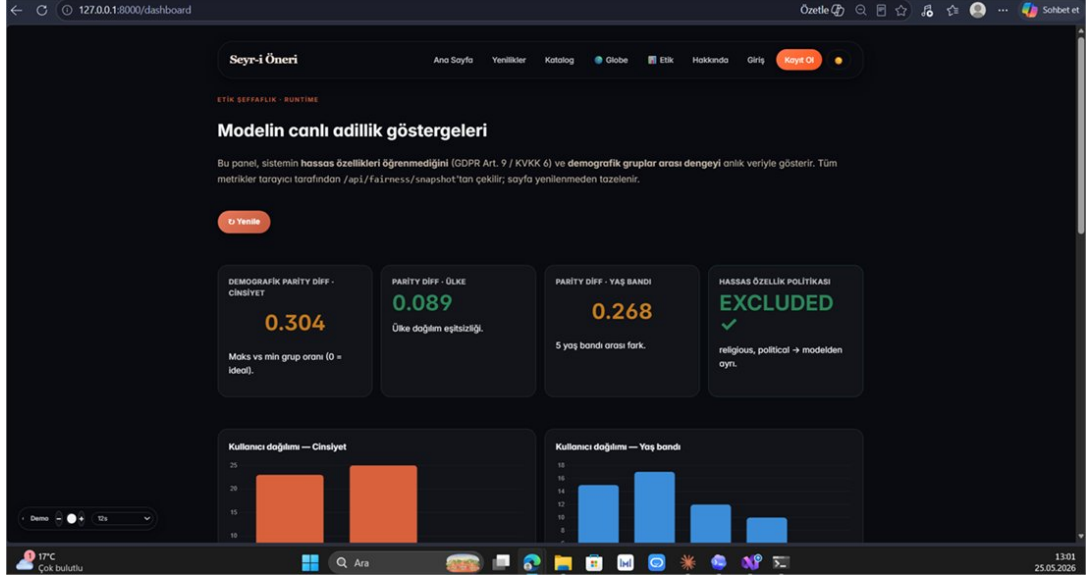


Figure 10: Ethics and fairness dashboard: demographic distributions, parity difference, and CLAR indicators.

9.4 Recommendation Page

The recommendation page is the centre of the system. Each product card contains category, score, reason labels, store information, satisfaction score, and a “Why this?” explanation button. The upper area gives meta information such as selected model, cognitive-load band, and cultural distance. The live-stream button uses SSE to show recommendations sequentially without refreshing the page.

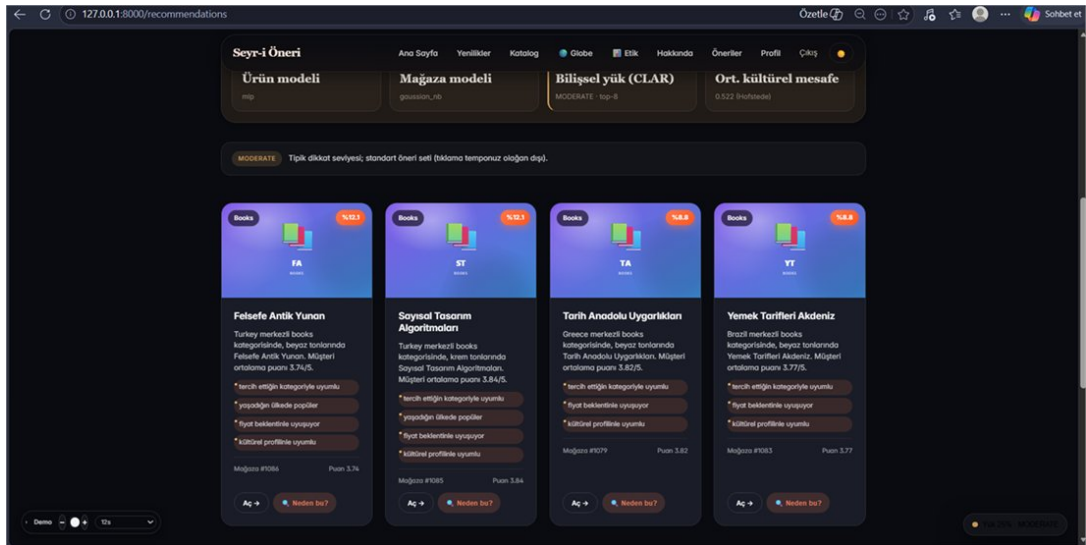


Figure 11: Personal recommendation page: hybrid score, reason labels, CLAR band, and JIT-XAI access.

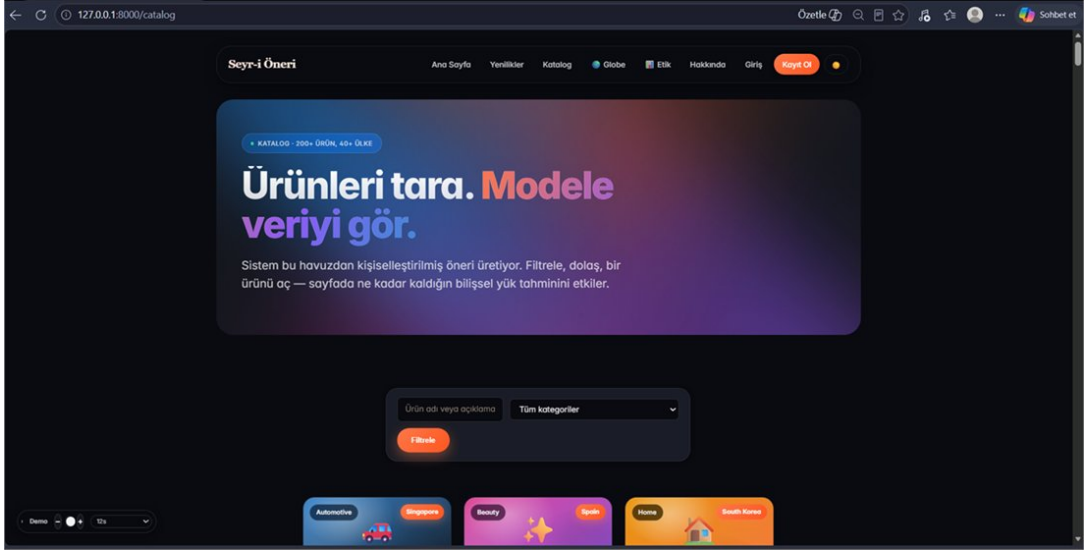


Figure 12: Catalogue page: 119 products, category filter, and deterministic SVG product visuals.

10 DISCUSSION

10.1 Interpretation of Main Findings

The most important finding of this study is that recommendation systems cannot be evaluated adequately through a single accuracy metric. CCE can produce neutral or slightly negative results in some classical ranking tasks. At first glance, this may appear to be a weakness; in fact, it increases the scientific maturity of the study. Cultural embedding does not automatically improve every problem. It produces value when the task is compatible with cultural context, transfer, interpretability, and diversity.

CLAR targets user load rather than accuracy. The reduction of average top-N from 8.00 to 5.47 shows that the system intentionally chooses to present fewer options in some sessions. This is a user-centred design response to the classical assumption that more recommendations are always better.

TDM does not erase the past. Instead, it weights historical interactions according to recency. The system can therefore remember user habits while reducing the risk of repeatedly recommending the same familiar products.

CUR adds a causal question to the recommendation system. Because the current data do not contain real A/B exposure logs, this layer currently uses synthetic propensity. Nevertheless, measuring uplift heterogeneity prepares the architectural foundation for future real experimental infrastructure.

JIT-XAI moves the model decision to the user. The fact that explanations can be generated with only a few milliseconds of additional p99 latency weakens the argument that explainability must be pushed into the background for performance reasons.

10.2 Limitations

Table 17: Limitations and mitigation strategies

Limitation	Effect	Mitigation / Future Work
Synthetic data	May not represent the full complexity of real user behaviour.	External validation with Retail-Rocket, Amazon Reviews, or a real pilot study.
Hofstede country level	Individual and within-country cultural diversity can be ignored.	Individual preference questionnaires or regional cultural subdimensions.
CUR synthetic treatment	No real recommendation-exposure log is available.	Production A/B exposure logging and randomised holdout.
CLAR calibration	Signal thresholds may be synthetic or literature-based.	Eye-tracking, user study, SUS, and NASA-TLX measurement.
UI CDN dependencies	Globe and Chart.js may be limited in offline settings.	Packaging dependencies as local bundles.
SQLite scale	Limited for multi-user production traffic.	PostgreSQL, async SQLAlchemy, and Redis cache.

10.3 Threats to Validity

Internal validity. Synthetic data-generation formulas partially determine the structure that the model will learn. Therefore, the experimental results are valid under the stated generation assumptions.

External validity. The MovieLens-100k test shows that some components are transferable; however, modules such as CCE and CLAR require country and behavioural context and cannot be tested directly on every public dataset.

Construct validity. Engagement is defined through dwell time and satisfaction score. Real purchase, return, or repeat-purchase signals could provide a stronger target variable.

Conclusion validity. Some experimental differences are small. For this reason, Wilcoxon testing, bootstrap confidence intervals, and reporting of negative findings are especially important.

10.4 Ethical Evaluation

The ethical contribution of the system is not limited to removing sensitive fields. The main contribution is that this decision is made reproducible and auditable at code level. The `MODEL_FEATURE_COLUMNS` constant does not contain sensitive fields; tests verify this invariant; the CI guard stops the build if the invariant is violated. This approach closes the gap between an “ethical statement” and “ethical implementation.”

In addition, JIT-XAI allows the user to receive an explanation at the moment of decision. The fairness dashboard makes demographic distributions visible. Together, these two elements support a recommendation system that is not only predictive, but also accountable.

11 CONCLUSION AND FUTURE WORK

11.1 Conclusion

Within the scope of this study, Seyr-i Öneri, an ethics-first hybrid retail recommendation system with cultural, cognitive, temporal, causal, and explainable layers, was designed and implemented. The system consists of a web application built with FastAPI and Jinja2, an SQLite-based data layer, scikit-learn/LightGBM models, SHAP explanation infrastructure, five innovation services, interactive interfaces, scientific experiment scripts, and automated tests.

The study shows that modern recommendation systems cannot be defended through accuracy alone. The correct question is not limited to “How many predictions did the model make correctly?” The system must also ask: “Did it present too many options to the user?”, “How current did it assume historical behaviour to be?”, “Did the recommendation actually change behaviour?”, “Did the user understand why this recommendation was shown?”, and “Were sensitive fields truly kept outside the model?”

The experimental findings show that the BASE LightGBM model performs strongly on the binary engagement task; that CCE is neutral in terms of accuracy but valuable for explanation and cultural-transfer reasoning; and that CLAR, TDM, CUR, and JIT-XAI provide measurable contributions in their own operational metrics. The system was verified with 80 automated tests and approximately 75 percent line coverage. As an accepted congress study, it was prepared for international presentation with a reproducible technical and academic foundation.

11.2 Future Work

Future work should proceed along five main directions:

1. **Real A/B exposure logs:** training the CUR layer with real treatment-control data instead of synthetic propensity.
2. **User study:** measuring the contributions of CLAR and JIT-XAI through SUS, NASA-TLX, NPS, and qualitative interviews.
3. **Public benchmark expansion:** external validation on RetailRocket, Amazon Reviews, and e-commerce behaviour logs.

4. **Production architecture:** PostgreSQL, Redis, asynchronous data access, observability, rate limiting, and CSRF protection.
5. **Cultural fairness:** measuring the effects of CCE beyond accuracy through cultural diversity, long-tail exposure, and cross-country cold-start behaviour.

In conclusion, Seyr-i Öneri is a compact but comprehensive reference implementation for recommendation systems that are culturally aware, cognitively gentle, causally honest, and capable of presenting explanations to users. Its value lies not in a single metric, but in the disciplined integration of scientific modelling, software architecture, reproducibility, and ethical constraints.

REFERENCES

- [1] British Congress. Vi international british congress information page. <https://www.britishcongress.co.uk/tr>, **2026**.
- [2] G. Hofstede, G. J. Hofstede, and M. Minkov. Cultures and organizations: Software of the mind, **2010**.
- [3] M. De Mooij. Consumer behavior and culture, **2019**.
- [4] J. Sweller. Cognitive load during problem solving, **1988**.
- [5] S. S. Iyengar and M. R. Lepper. When choice is demotivating, **2000**.
- [6] D. Bollen, B. P. Knijnenburg, M. C. Willemsen, and M. Graus. Understanding choice overload in recommender systems, **2010**.
- [7] S. R. Kunzel, J. S. Sekhon, P. J. Bickel, and B. Yu. Metalearners for estimating heterogeneous treatment effects, **2019**.
- [8] S. Athey and G. Imbens. Recursive partitioning for heterogeneous causal effects, **2016**.
- [9] E. Diemert, A. Betlei, C. Renaudin, and M. R. Amini. A large scale benchmark for uplift modeling, **2018**.
- [10] S. M. Lundberg and S.-I. Lee. A unified approach to interpreting model predictions, **2017**.
- [11] B. Goodman and S. Flaxman. Eu regulations on algorithmic decision-making and a right to explanation, **2017**.
- [12] Y. Zhang and X. Chen. Explainable recommendation: A survey and new perspectives, **2020**.
- [13] F. Ricci, L. Rokach, and B. Shapira. Recommender systems handbook, **2022**.
- [14] Y. Koren, S. Rendle, and R. Bell. Advances in collaborative filtering, **2022**.
- [15] S. Zhang, L. Yao, A. Sun, and Y. Tay. Deep learning based recommender system: A survey, **2019**.
- [16] B. McSweeney. Hofstede model of national cultural differences and their consequences, **2002**.
- [17] Y. Ding and X. Li. Time weight collaborative filtering, **2005**.

- [18] J. Vinagre, A. M. Jorge, and J. Gama. Exploitation of time in collaborative filtering, **2015**.
- [19] M. D. Ekstrand, R. Burke, and F. Diaz. Fairness and discrimination in recommendation and retrieval, **2019**.
- [20] D. Pessach and E. Shmueli. A review on fairness in machine learning, **2023**.
- [21] P. Covington, J. Adams, and E. Sargin. Deep neural networks for youtube recommendations, **2016**.
- [22] M. F. Dacrema, P. Cremonesi, and D. Jannach. Are we really making much progress?, **2019**.
- [23] H. Ahmadian Yazdi, S. M. R. Seyyedi, and A. Asghari Ivani. Cross-cultural and personality factors in recommender systems, **2019**.
- [24] R. Hu and P. Pu. User perception of personality-based recommender systems, **2010**.
- [25] B. P. Knijnenburg, M. C. Willemsen, Z. Gantner, H. Soncu, and C. Newell. Explaining the user experience of recommender systems, **2012**.
- [26] C. Anderson. The long tail, **2006**.
- [27] A.-L. Barabasi. The origin of bursts and heavy tails in human dynamics, **2005**.
- [28] C. Liu, R. W. White, and S. Dumais. Understanding web browsing behaviors through dwell time, **2010**.

A Appendix-1: Technical Component Inventory

This appendix summarises the source-code-level technical inventory of the software components explained in the main body of the report. The purpose is to show, in a consolidated form, which main modules constitute the developed system and what responsibility each module has within the research prototype.

Table 18: Source-code modules and responsibilities

File / Module	Responsibility
app/main.py	FastAPI application, middleware, static mount, and router inclusion.
app/config.py	Path constants, secret-key loading, and artefact locations.
app/constants.py	Ethical feature set, CCE feature set, feature flags, and form fields.
app/db.py	SQLite schema and connection management.
app/controllers/web.py	Pages, API endpoints, and singleton service binding.
app/services/recommendation_service.py	Hybrid recommendation orchestration and integration of the five innovation layers.
app/services/cultural_embedding.py	Hofstede six-dimensional cultural vector and cultural distance.
app/services/cognitive_load.py	CLAR behavioural signals, load score, and adaptation table.
app/services/temporal_memory.py	Exponentially decayed behavioural memory and familiarity penalty.
app/services/causal_uplift.py	T-learner uplift modelling and re-ranking.
app/services/explainability.py	SHAP-based JIT-XAI explanations.
scripts/train_scientific.py	Multiple baselines, cross-validation, bootstrap, and Wilcoxon experiments.
scripts/ablate_innovations.py	Ablation-lattice analysis of the five innovations with separate metrics.
scripts/external_validity_movielens.py	MovieLens-100k external-validity test.

B Appendix-2: Selected Code Listings

This appendix presents selected code snippets that technically support the academic claims of the system. The snippets summarise the implementation of the ethical feature contract, temporal decay, and causal uplift components.

B.1 Ethical Feature Contract

```
1 MODEL_FEATURE_COLUMNS = [  
2     "enc_country",  
3     "enc_city",  
4     "enc_county",  
5     "enc_gender",  
6     "z_age",  
7     "z_spent_on_product_page",  
8     "enc_the_category_of_products_it_receives",  
9     "enc_color_of_the_product",  
10    "is_product_expensive_int",  
11    "enc_shopping_frequency",  
12 ]  
13  
14 SENSITIVE_FIELDS = [  
15     "religious_orientation",  
16     "target_religious_orientation",  
17     "political_orientation",  
18 ]
```

Code Listing 4: Production feature set without sensitive fields

B.2 TDM Weight

```
1 def _weight(self, age_days: float) -> float:  
2     if age_days < 0:  
3         age_days = 0.0  
4     return math.exp(-self._decay_rate * age_days)
```

Code Listing 5: Ebbinghaus exponential decay weight

B.3 CUR Uplift Prediction

```
1 def predict_uplift(self, X: pd.DataFrame) -> np.ndarray:  
2     if not self._fitted:  
3         return np.zeros(len(X), dtype=np.float32)  
4     if list(X.columns) != self._feature_columns:  
5         X = X.reindex(columns=self._feature_columns, fill_value=0.0)  
6     mu1 = self.mu1.predict(X)  
7     mu0 = self.mu0.predict(X)  
8     return np.asarray(mu1 - mu0, dtype=np.float32)
```

Code Listing 6: T-learner uplift prediction

C Appendix-3: Database Schema

This appendix gives the SQLite schema outline that forms the core data model of the prototype application. The schema covers user profiles, the product catalogue, and user-product interaction records, thereby supporting both experimental evaluation and reproducible live demonstration flow.

```
1 CREATE TABLE users (  
2     id INTEGER PRIMARY KEY AUTOINCREMENT,  
3     full_name TEXT NOT NULL,  
4     email TEXT NOT NULL UNIQUE,  
5     password_hash TEXT NOT NULL,  
6     age INTEGER NOT NULL,  
7     gender TEXT NOT NULL,  
8     country TEXT NOT NULL,  
9     city TEXT NOT NULL,  
10    county TEXT NOT NULL,  
11    shopping_frequency TEXT NOT NULL,  
12    religious_orientation TEXT NOT NULL,  
13    target_religious_orientation TEXT NOT NULL,  
14    preferred_category TEXT NOT NULL,  
15    preferred_color TEXT NOT NULL,  
16    political_orientation TEXT NOT NULL,  
17    prefers_expensive INTEGER NOT NULL DEFAULT 0,  
18    avg_page_seconds REAL,  
19    created_at TEXT NOT NULL,  
20    updated_at TEXT NOT NULL  
21 );  
22  
23 CREATE TABLE products (  
24     product_store_key TEXT PRIMARY KEY,  
25     product_name TEXT NOT NULL,  
26     category TEXT NOT NULL,  
27     color TEXT NOT NULL,  
28     is_expensive INTEGER NOT NULL,  
29     country TEXT NOT NULL,  
30     company_id INTEGER NOT NULL,  
31     store_id INTEGER NOT NULL,  
32     product_id INTEGER NOT NULL,  
33     popularity INTEGER NOT NULL,  
34     avg_satisfaction REAL NOT NULL,  
35     description TEXT NOT NULL  
36 );  
37  
38 CREATE TABLE user_events (  
39     id INTEGER PRIMARY KEY AUTOINCREMENT,  
40     user_id INTEGER NOT NULL,  
41     product_store_key TEXT NOT NULL,  
42     product_name TEXT NOT NULL,  
43     view_seconds REAL,  
44     satisfaction_score INTEGER,  
45     created_at TEXT NOT NULL,  
46     FOREIGN KEY(user_id) REFERENCES users(id)  
47 );
```

Code Listing 7: Seyr-i Oneri SQLite schema outline

D Appendix-4: Congress Acceptance Information

The English abstract of the study, entitled “Design and Implementation of a Culturally Grounded, Cognitively Aware, and Causally Honest Hybrid Retail Recommendation System: Seyr-i Öneri”, was evaluated within VI. International British Congress on Interdisciplinary Scientific Research & Practices and accepted under **Ref: AC-06060** on **28.05.2026** for oral presentation and proceedings inclusion. The congress is organised on 18–21 June 2026 with London, United Kingdom as the host location [1].

Table 19: Summary of acceptance record and congress information

Field	Information
Acceptance reference	AC-06060
Acceptance date	28.05.2026
Presentation type	Oral presentation
Congress	VI. International British Congress on Interdisciplinary Scientific Research & Practices
Place and date	London, United Kingdom, 18–21 June 2026
Institution	OSTIM Technical University
Author	Samed Alp ARSLAN

CURRICULUM VITAE

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RESEARCH INTERESTS

Samed Alp ARSLAN is a Software Engineering student at OSTIM Technical University. His academic and technical interests include scalable web-based software development with the .NET architecture and ASP.NET Core MVC framework, relational database design and optimization, and the integration of artificial intelligence technologies into business processes. He has worked on projects involving large language models (LLMs), generative artificial intelligence, deep learning algorithms such as CNN and RNN/LSTM, real-time data processing, and computer vision. His work particularly focuses on user-oriented information organization and task management architectures within enterprise resource planning (ERP) and customer relationship management (CRM) systems.